

Econ 101A

Section 3 - Solutions

Bruno Smaniotto and Anson Tam*

January 31, 2025

1 Utility Functions

Definition: A utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$ if, for every pair of points x and y in X , $x \succsim y$ if and only if $u(x) \geq u(y)$. In other words, if x has higher utility than y , you prefer x to y .

- **Theorem (Existence):** If preference relation $\succsim$ is rational and continuous on set X , there exists a continuous utility function $u : X \rightarrow \mathbb{R}$ that represents it.
- **Non-uniqueness:** The utility function representing preferences is not unique. If $u(x)$ represents preferences $\succsim$ and f is a strictly increasing function, then $f(u(x))$ represents $\succsim$ as well. For most of the time we only care about the **ordinal** property of utility function.

2 Univariate Optimization

- Optimization is about finding optimums, i.e. maximums and minimums, of *scalar* functions.
- Recall that maximizing a function is equivalent to minimizing its negative. Thus:

$$\operatorname{argmax}_{x \in X} f(x) = \operatorname{argmin}_{x \in X} -f(x)$$

where $\operatorname{argmax}_{x \in X} f(x)$ ($\operatorname{argmin}_{x \in X} f(x)$) gives the solutions to the problem $\max_{x \in X} f(x)$ ($\min_{x \in X} f(x)$).

- A stationary point, or critical point, is a point at which a function is neither going up nor down - i.e. an x_0 at which $f'(x_0) = 0$. In practice finding an optimum usually means finding a stationary point, but note that being a stationary point is neither a necessary nor a sufficient condition for being an optimum, since we could have a corner solution.
- Considering only interior solutions (ignoring corner solutions), what must be true at a local maximum(minimum)?
 - FOC: The function is at a stationary point: $f'(x^*) = 0$
 - SOC: On either side of the stationary points, the function must be going down(up): $f''(x^*) < 0$ ($f''(x^*) > 0$).

*These notes are a consolidation of the notes of Jianmeng Lyu, Dan Acland, Ivan Balbuzanov, Mariana Carrera, Justin Gallagher, Anne Karing, Matt Leister, Sam Leone, Nicholas Li, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Katalin Springel, Jonas Tungodden, David Wu, Sam Wang, and Matteo Saccarola.

2.1 Necessary and Sufficient Conditions in the Context of Optimization

- FOC is a necessary condition for an **interior local** optimum.
- SOC is a necessary condition for an **interior local** optimum.
- Neither, taken alone, is a sufficient condition for an **interior local** optimum.
- FOC and SOC together are necessary and sufficient conditions for an **interior local** optimum.

2.2 Optimization of Concave Functions

Usually, to find global maximum (if it exists) of a function in a given range, we need to find all local maxima and pick the largest one. With concavity, it suffices to find the only stationary point with FOC.

- Definition: Function f from a convex set $C \subset R^n$ to R is concave (convex) if

$$f(tx + (1-t)y) \geq (\leq) tf(x) + (1-t)f(y)$$

for all $x, y \in C$ and for all $t \in [0, 1]$.

- A more useful characterization: suppose $f(x) : R \rightarrow R$ is twice differentiable, then f is concave iff $f'' \leq 0$ for **all** x .
- A useful result: Consider a **twice-differentiable concave** function over $C \subset R^n$. If x_0 satisfies FOC, then it is a global maximum.
 - Not applicable to cases where f'' is not globally identified (e.g. $f(x) = -|x|$, $x \in [-1, 1]$)
 - This is a **sufficient** condition for global maximum. Corner solutions are possible if no points satisfy FOC in the given range (e.g. $f(x) = -x^2$, $x \in [1, 2]$. What if $x \in [-1, 0) \cup (0, -1]$?)

3 Exercises

3.1 Problem 1 from PS1, Spring 2014

Consider the following maximization problem: $\max_x f(x; x_0) = \exp(-(x - x_0)^2)$

1. Write down the first order conditions for this problem with respect to x (notice that x_0 is a parameter).
 - The first order condition with respect to x is $\partial f(x; x_0)/\partial x = 0$.
 - Plugging in the function in question yields $-2(x - x_0) * \exp(-(x - x_0)^2) = 0$.
2. Solve explicitly for the x^* that satisfies the first order conditions.
 - Solving the first order condition yields $x^* = x_0$
3. Now compute the second order conditions at x^* . Is the stationary point that you found in (2) a maximum? Why or why not?
 - Computing the second order conditions, we find that

$$\partial^2 f(x; x_0)/\partial x^2 = (-2 + 4(x - x_0)^2) * \exp(-(x - x_0)^2)$$

- Plugging in $x^* = x_0$ we find that the second derivative is in fact negative.
 - Furthermore, since $x^* = x_0$ is the only point that satisfies the first order condition, it follows that it is the only local maximum. Since the function f tends to zero as x tends to either plus or minus infinity, it follows that the point $x^* = x_0$ is a global maximum.
4. Is the function f concave in x ?
 - No, it is not concave.
 - Recall that the second derivative of $f(x; x_0)$ with respect to x must be non-positive for concavity.
 - However, we found that this derivative was equal to:
 $(-2 + 4(x - x_0)^2) \times \exp(-(x - x_0)^2)$.
 - For example, whenever $x - x_0 > \frac{\sqrt{2}}{2}$ or $x - x_0 < -\frac{\sqrt{2}}{2}$, the term is positive.

3.2 Problem 2 from PS1, Spring 2017

Consider the following maximization problem: $\max_x f(x; p) = (p - x)(x - 1)$. p is a parameter.

1. Write down FOC with respect to x . Solve explicitly for x^* that satisfies FOC.
Answer: FOC: $-2x + p + 1 = 0$, $x^* = \frac{p+1}{2}$.
2. Compute SOC. Is the stationary point that you found in part 2 a local maximum?
Answer: SOC: $f''(x^*) = -2 < 0$, so it's a local maximum.
3. Is function f concave in x ? Is x^* a global maximum?
Answer: Yes, because $f''(x) = -2 \forall x \in R$. Thus x^* is a global maximum $\forall x \in R$.
4. Comparative statics: compute dx^*/dp .
Answer: Just look at the expression of $x^* = \frac{p+1}{2}$: $dx^*/dp = 1/2$.
5. Now we want to verify Envelope Theorem (Cecile will prove this theorem in next class):

- Plug x^* back in function f and compute $f(x^*(p); p)$ (so now there's no x in the expression of f). Then, take the derivative with respect to p .

Answer: Assume $g(p) = f(x^*(p); p)$, then

$$g(p) = \left(p - \frac{p+1}{2}\right) \left(\frac{p+1}{2} - 1\right) = \left(\frac{p-1}{2}\right)^2$$

$$\frac{dg(p)}{dp} = \frac{p-1}{2}$$

- Let's try a second way: first, treat f as a function of p and directly write down FOC w.r.t. p (treat x as a parameter now). Then, plug x^* in FOC. Compare your answer to the previous part.

Answer: $\frac{\partial f(x; p)}{\partial p} = x - 1$. Plug in $x^* = \frac{p+1}{2}$: $\frac{\partial f(x; p)}{\partial p}|_{x=x^*} = \frac{p-1}{2}$, same as above.